RK818/816 Fuel Gauge Print Information Introduction

ID: RK-SM-YF-003

Release Version: V1.1.0

Release Date: 2021-06-08

Security Level: □Top-Secret □Secret □Internal ■Public

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Preface

Overview

The RK818/RK816 integrates fuel gauge and charge functionalities, specifically battery capacity detection and battery charging management. The implementation of these features is highly similar across both chips. This document uses the RK818 as the primary example to introduce key information printed by the Driver at various stages. The log output for the RK816 is similar and can be referenced directly; therefore, a separate document is not provided.

Product Version

Chip Models	Kernel Version
RK818, RK816	Linux 4.4, Linux 4.19

Intended Audience

This document (this guide) is primarily intended for the following engineers:

- Technical Support Engineers
- Software Development Engineers

Revision History

Version	Author	Date	Description
V1.0.0	Chen Jianhong	2018-05-28	Initial version
V1.1.0	Chen Jianhong	2021-06-08	Added support for kernel 4.19

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1. Charger/OTG Detection

1.1 Driver Files

```
drivers/power/rk818_charger.c
```

This Driver primarily implements detection of charger/OTG plug-in/plug-out events and configuration of charging current. All log messages use "rk818-charger: " as a prefix for easy identification. The terms "ac", "usb", "dc", and "otg" appearing in the logs represent different devices, where 1 indicates a connected state and 0 indicates a disconnected state. Note that the PMIC itself lacks the capability to detect charger/OTG plug-in/plug-out events; therefore, charger type detection relies entirely on USB notification chain messages.

1.2 probe Phase

DC Charger Registration Status:

```
rk818-charger: support dc
rk818-charger: not support dc
```

Register the notification chain for the Type-C port charger:

```
rk818-charger: register typec extcon evt notifier
```

Register the notification chain for the traditional USB port charger:

```
rk818-charger: register bc evt notifier
```

Connection status of each device at the end of probe:

```
rk818-charger: ac=1, usb=0, dc=0, otg=0
```

Driver Version:

```
rk818-charger: driver version: 2.0
```

1.3 Running Phase

Charger/OTG device plug-in and plug-out messages from the USB notification chain:

```
rk818-charger: receive bc notifier event: DISCNT    // Charger removed
rk818-charger: receive bc notifier event: USB       // PC charging inserted
rk818-charger: receive bc notifier event: AC        // Standard charger inserted
rk818-charger: receive bc notifier event: CDP1.5A   // CDP charger inserted
rk818-charger: receive bc notifier event: UNKNOWN   // Unrecognized charger inserted
rk818-charger: receive bc notifier event: OTG ON    // OTG insertion
rk818-charger: receive bc notifier event: OTG OFF   // OTG unplugged
rk818-charger: detect dc charger in                 // DC inserted
rk818-charger: detect dc charger out                // DC unplugged
```

The PMIC itself cannot determine the charger type, but it can detect whether a charging device is inserted:

```
rk818-charger: pmic: plug out
rk818-charger: pmic: plug in
```

5V power supply variation during OTG device insertion and removal:

```
rk818-charger: disable otg5v
rk818-charger: enable otg5v
```

Device and current information are updated after each charger/OTG plug-in or plug-out event:

```
rk818-charger: ac=1 usb=0 dc=0 otg=0 v=4200 chrg=1000 input=1800 virt=0
```

1.4 Suspend Phase

Changes in the 5V power supply for the OTG device during suspend:

```
rk818-charger: suspend: otg 5v on
rk818-charger: suspend: otg 5v off
```

1.5 Shutdown Phase

Connection status of each device during shutdown:

```
rk818-charger: shutdown: ac=1 usb=0 dc=0 otg=0
```

2. Battery Fuel Gauge

2.1 Driver File

```
drivers/power/rk818_battery.c
drivers/power/rk818_battery.h
```

This Driver primarily implements the fuel gauge functionality, providing a set of routines for statistics on battery capacity information. All log messages are prefixed with "rk818-bat: " for easy identification.

2.2 probe Phase

Upon the first power-on after connecting the battery, the message "first on" is displayed:

```
rk818-bat: first on: dsoc=24, rsoc=24 cap=960, fcc=4000, ov=3840
```

When an abnormal shutdown (e.g., continuous power consumption after a system hang) causes a coulomb counter anomaly, a forced calibration of the coulomb counter is performed upon the next boot:

```
rk818-bat: system halt last time... cap: pre=2400, now=120
```

When U-Boot has already initialized the fuel gauge, the kernel fuel gauge Driver can skip certain initialization steps to prevent redundant initialization:

```
rk818-bat: initialized yet.
```

Initial state of the coulomb counter during the probe phase:

```
rk818-bat: dsoc=32 cap=1000 v=3780 ov=3900 rv=3890 min=25 psoc=32 pcap=1000
```

Fuel Gauge Version:

```
rk818-bat: driver version 7.1
```

2.3 Running Phase

Each time the battery level changes, the Driver reports the status to the framework with the following log output: the first line indicates the real-time status of each parameter; the second line shows the parameter status at boot initialization, primarily used for debugging.

```
rk818-bat: changed: dsoc=22, rsoc=24, v=3820, ov=3770 c=1018, cap=960, f=4000, st=cc cv,
hotdie=0
rk818-bat: dl=10, rl=12, v=3670, halt=0, halt_n=0, max=0, init=0, sw=0, calib=0, below0=0,
force=0
```

2.4 Suspend Phase

After the system enters deep sleep, if the battery voltage drops below the shutdown threshold due to an extended standby period, a PMIC wake-up Interrupt is generated, followed by system shutdown:

```
rk818-bat: lower power yet, power off system! v=3350, c=-125, dsoc=0
```

2.5 Shutdown Phase

Display relevant critical information during shutdown:

```
rk818-bat: shutdown: dl=0 rl=2 c=-1220 v=3460 cap=88 f=4000 ch=1 n=0 mode=1 rest=128
Printout meaning:<Displaysoc> <Realsoc> <Current> <Voltage> <Remaining Capacity> <Full Charge
Capacity> <Is therecharger> <Ignore the rest.....>
```

3. About the RK816 Fuel Gauge

The functionality of the RK816 fuel gauge is largely similar to that of the RK818. Modifications related to charger identification and the fuel gauge are consolidated in:

```
drivers/power/rk816_battery.c
```

The log output is largely similar to that of the RK818, using the prefix "rk816-bat:" for identification. Therefore, for RK816-related information, refer to the RK818 documentation above.